

JEAN KIM

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EXPERIENCE

Polymeric Materials Engineering Intern @NASA ----- Jun - Aug 2026 (3 mos)
NASA Glenn Research Center • Cleveland, Ohio

- Characterized degradation of aerospace wire insulation via TGA, ASTM E595 outgassing, vacuum-oven exposure, FTIR, and Raman; resolved fiberglass, PTFE, and PVA binder chemistries and built a standardized testing and documentation procedure that made results usable by non-materials audiences.
- First-author publication: extended prior polymer color-analysis machine learning research to complex materials under extreme-heat environments, implementing new models for 15% higher accuracy; authored the resulting manuscript as first author for submission to a peer-reviewed journal. Also named co-author on a lunar habitation composites study, contributing DSC and outgassing data.
- Delivered a materials failure assessment on thruster structural integrity to NASA and industry-partner engineers; empirical characterization data drove a change in project scope and launched a follow-on effort that I now lead.
- Independently operated TGA, DSC, FTIR, Raman, ASTM E595 outgassing, LIBS, and rheological (thermoplastic viscosity) instrumentation, with assisted experience on SEM, SEM-EDS, C-Therm, and injection molding; interpreted chemical structures of industry-grade polymers (PEEK, PAEK, PBI, PTFE, PVA, PI) and identified organic compounds directly from spectra.
- Built a trade study ranking alternative polymers by cost, manufacturability, and outgassing performance; authored SOPs for TGA, outgassing, and specimen photography, and coordinated external testing and manufacturer communication on material chemistry and qualification methods.

Summer Research Intern @AI Corporate Training Center ----- Jul - Aug 2025 (2 mos)
Seoul National University • Seoul, South Korea

- Research on next-generation computer memory under the mentorship of Professor Soojung Ryu.
- Key research findings: Comparative LPDDR/GDDR memory analysis for device applications, Integrated Accelerator system evaluation, Chip-to-Chip interconnect design optimization.

Summer Camp Manager @Seoul Foreign School ----- Jun - Jul 2024 (2 mos)
Seoul Foreign School • Seoul, South Korea

- Supervised a team of 30+ counselors, demonstrating strong multitasking abilities and leadership. Collaborated effectively in a team environment by providing conflict resolution and establishing clear communication protocols across diverse groups. Previously a counselor for June-Jul 2023 (2 mos)

EDUCATION

Bachelor of Science **Materials Science & Engineering** ----- Expected Jun 2027
University of Washington • Seattle, WA • GPA: 3.44

- Lab Experiments: XRD and SEM Crystallography | Electrical / Optical Property Characterization | Polymers Characterization | Ceramic / Metals Processing | 2D and Nano-materials Characterization

International Baccalaureate Diploma ----- Aug 2018 - May 2023
Seoul Foreign High School • Seoul, South Korea

PROJECTS & LEADERSHIP ACTIVITIES

Lead Ground Penetrating Radar Engineer • Husky Satellite Lab (HSL) ----- March 2026 - Present
University of Washington • Seattle, WA

- Developing and designing the ground penetrating radar (GPR) payload for HS-3's mission to detect and map habitable lava tubes under the lunar surface.

Project Lead • Boeing CFRP Wing Spoiler Quality Recovery Project ----- Sep - Dec 2025 (3 mos)
University of Washington • Seattle, WA

- Led team of 5 in advanced composite materials quality analysis project, implementing RCA and NDT testing methodologies to resolve manufacturing defects. Developed acceptance criteria, presented result to Boeing engineers.

CORE COMPETENCIES

- **Material Characterization:** TGA, DSC, FTIR, Raman, LIBS, XRD, SEM/EDS, Rheology
- **Programming:** Java (proficient), Python (learning), MatLab (learning), Ansys (learning)
- **Software:** MS Word, Excel and Power Point | Adobe Photoshop, Illustrator

AWARDS

From University of Washington:

- James I. Mueller Scholarship Recipient 25-26'
- Purple & Gold 4 Years Scholarship Recipient
- Dean's List: Aut 23' • Spr 25' • Aut 25'